PyGameMaker — Tutorial 15: Fruit Fusion

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Fruit Fusion: Catch and Merge!** (6 pages, about 20-25 minutes). Keep the tutorial open on one half of your screen. Tick each box when the step is done, and check the **You should see** line at the end of every phase. If something does not match, look at **Stuck?** before you call your teacher.

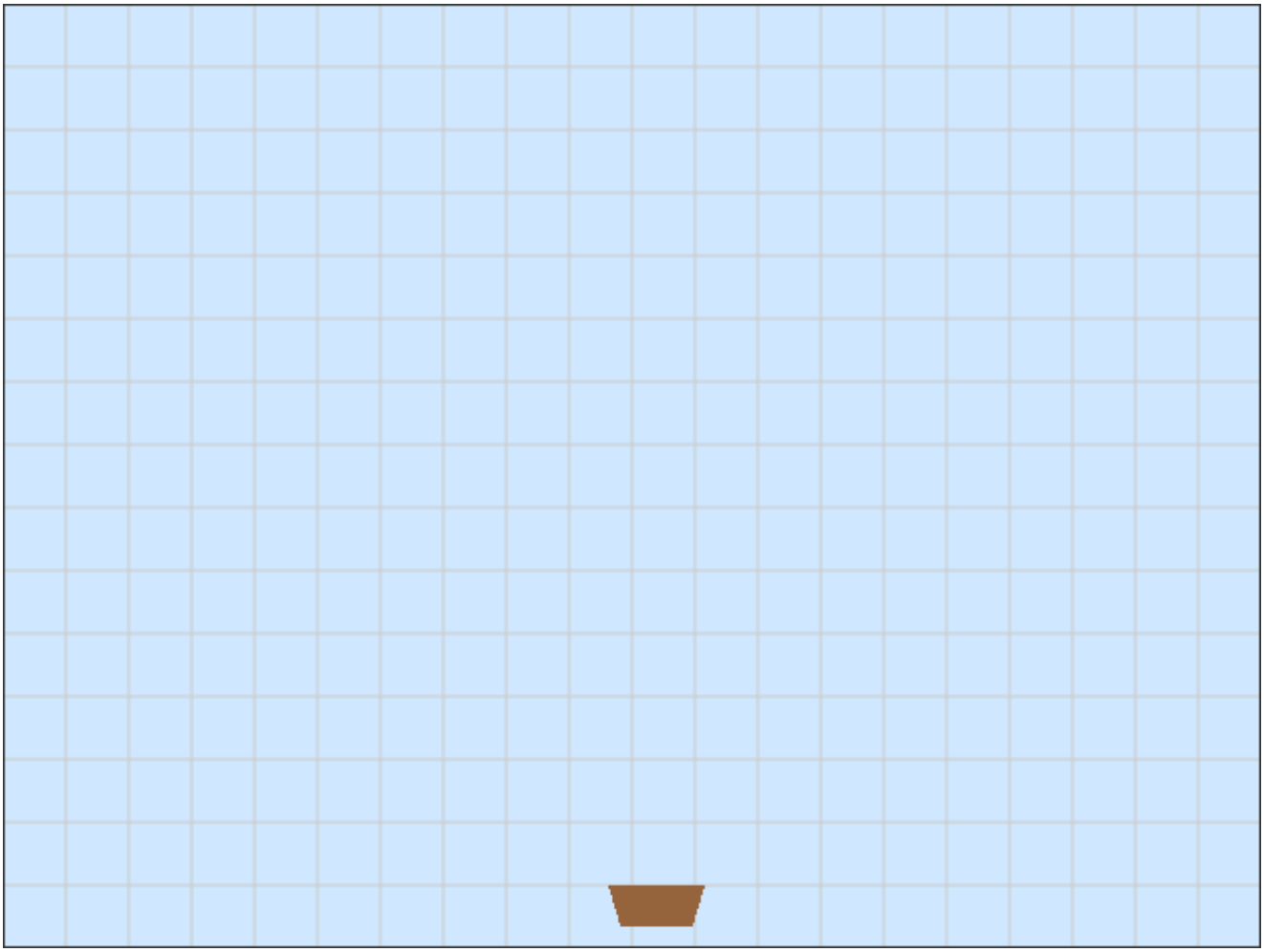
What You Will Make

A basket that always holds one fruit, starting with a cherry. Cherries, strawberries and oranges fall from the sky. Catch a fruit that matches what you are holding and it fuses into the next, bigger fruit. Fuse all the way up to a watermelon and you win!

Phase 1: Moving Basket

- [ ] **1.** Make a new project FruitFusion.
- [ ] **2.** Create a sprite spr_basket_empty (56x40, a brown bowl shape) and an object obj_player with that sprite.
- [ ] **3.** Rename the project's starting room (room0) to room_main, set its **width to 640** and **height to 480** in its properties, and place obj_player near the bottom center.
- [ ] **4.** In obj_player: **Keyboard: Left Arrow (held)** with *Set horizontal speed to -5*; **Right Arrow (held)** with *5*; **Keyboard: No key** with *Stop movement*.

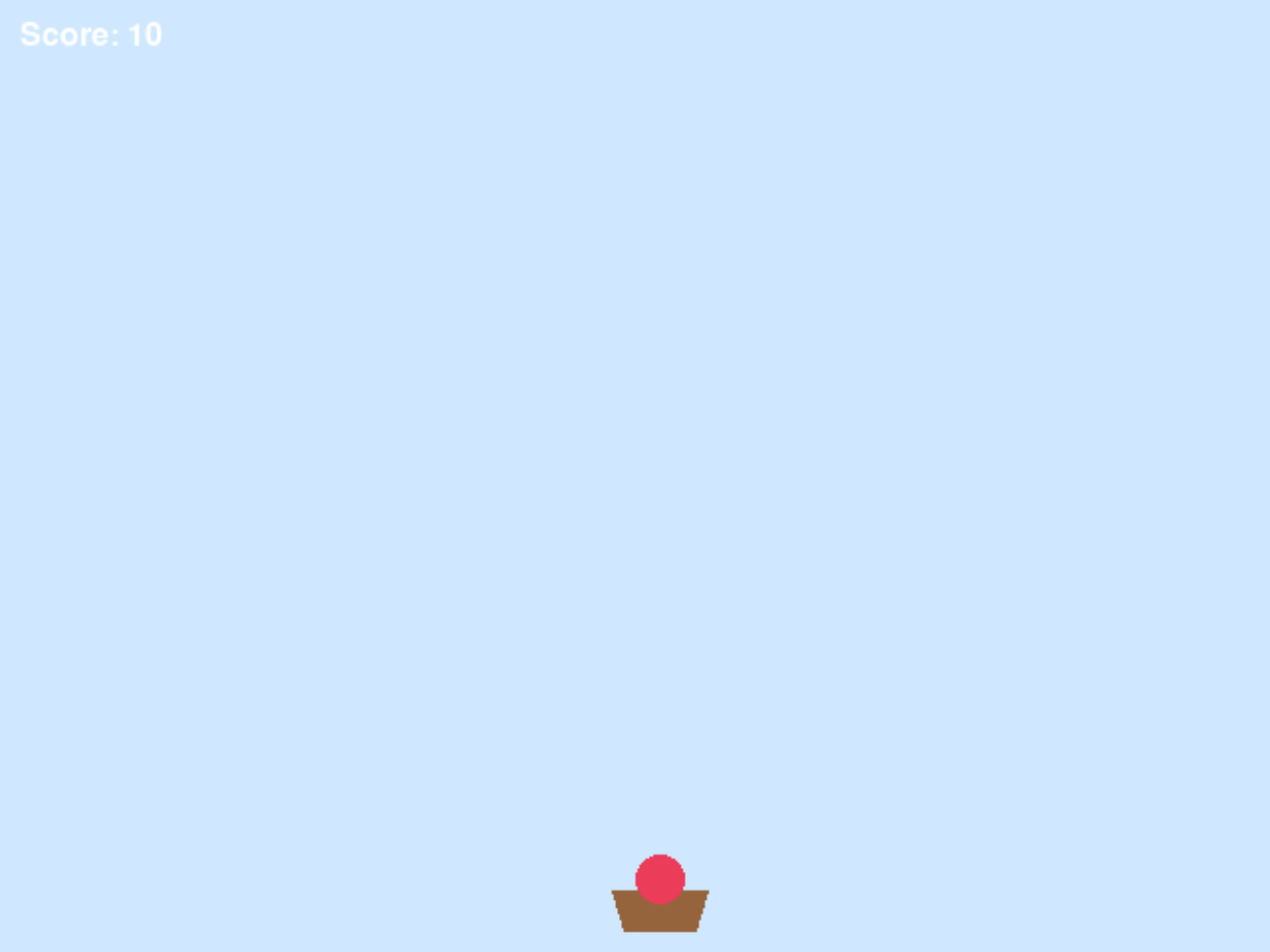
You should see: press **F5**. Left and Right move the basket, and it stops when you let go.



Phase 2: First Fusion

- [ ] **5.** Create a sprite spr_fruit_cherry (20x20, a red circle) and an object obj_fruit_cherry with a **Create** event (*Set vertical speed 3*) and an **Outside Room** event (*Destroy this instance*). **Outside Room** is not its own block — in the Events category, drag out **Other event:** and pick "Outside Room" from its dropdown.
- [ ] **6.** Create obj_spawn_cherry (no sprite) with a **Create** event (*Set alarm 0 to 90*) and an **Alarm 0** event (*Create instance of obj_fruit_cherry at x: random integer from 0 to 620, y: 0*, then *Set alarm 0 to 90* again). Place one in room_main. (620 = room width 640 minus the cherry's own width 20, so it never spawns hanging off the edge.)
- [ ] **7.** Change obj_player's sprite to a new sprite spr_basket_cherry (a bowl with a cherry in it) — the basket now starts out already holding a cherry.
- [ ] **8.** In obj_player: **Create** event sets *score to 0* and *variable held_level to 1*; **Draw** event does *Draw score at x 10, y 10*.
- [ ] **9.** Create a sprite spr_basket_strawberry (56x40, a bowl with a strawberry in it) — you'll need it in the next step.
- [ ] **10.** Add **Collision with obj_fruit_cherry**: *If held_level equals 1 then* [*set held_level to 2*, *add 10 to score*, *set sprite to spr_basket_strawberry*] *Otherwise* [*add 1 to score*]; then *Destroy other instance*.

You should see: press **F5**. Catch a falling cherry — the basket turns into a strawberry and the score jumps by 10.

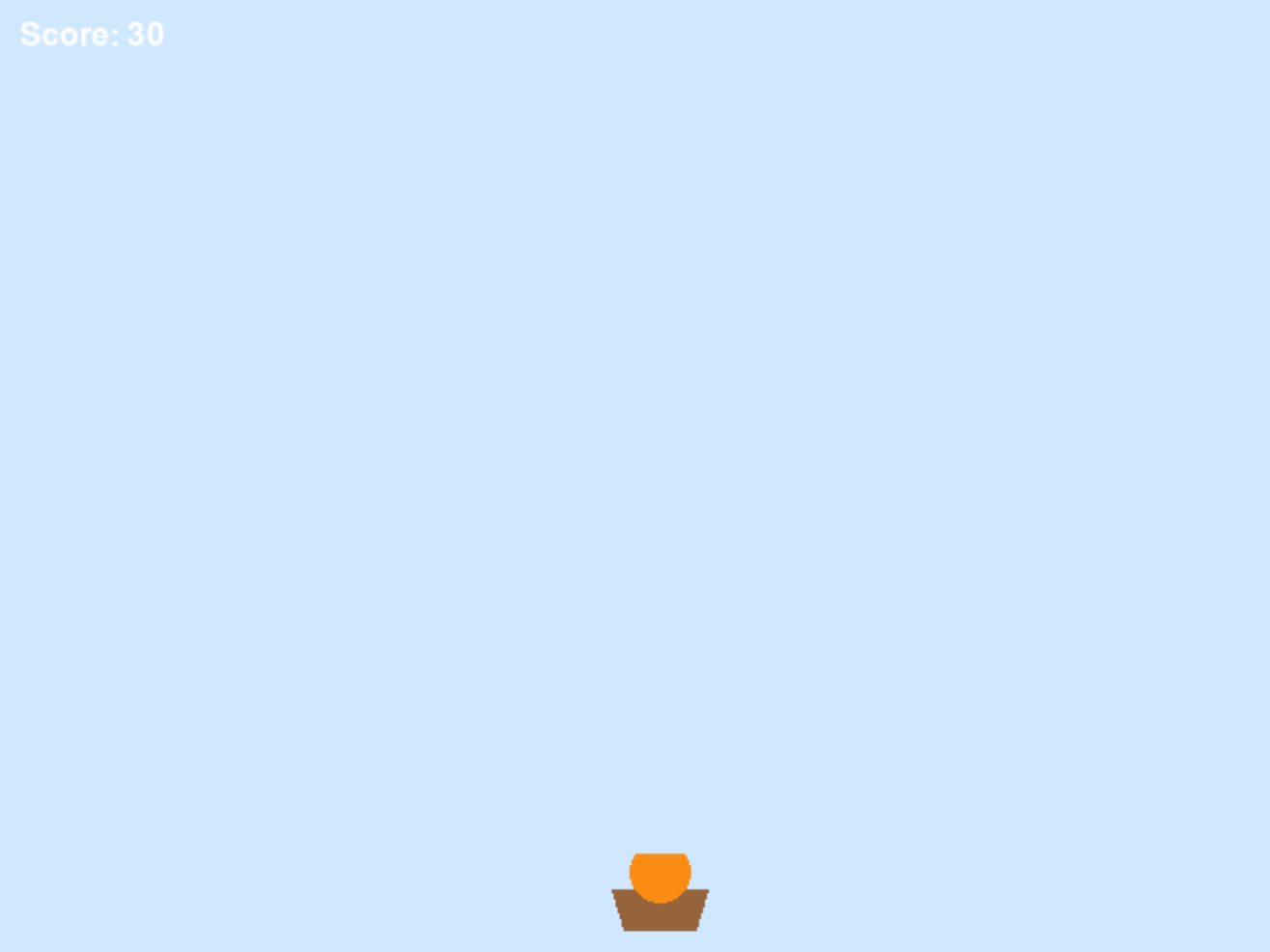


Phase 3: Second Fusion

This phase repeats the exact same recipe as Phase 2, with "cherry" changed to "strawberry" and the numbers changed from 1/2 to 2/3.

- [ ] **11.** Create a sprite spr_fruit_strawberry (26x26, a pink-red circle) and an object obj_fruit_strawberry, built exactly like obj_fruit_cherry.
- [ ] **12.** Create obj_spawn_strawberry (alarm 0 to 120 this time; random x from 0 to 614, since the strawberry is 26 wide) and place one in room_main.
- [ ] **13.** Create a sprite spr_basket_orange (56x40, a bowl with an orange in it) — you'll need it in the next step.
- [ ] **14.** Add **Collision with obj_fruit_strawberry** on obj_player: *If held_level equals 2 then* [*set held_level to 3*, *add 20 to score*, *set sprite to spr_basket_orange*] *Otherwise* [*add 1 to score*]; then *Destroy other instance*.

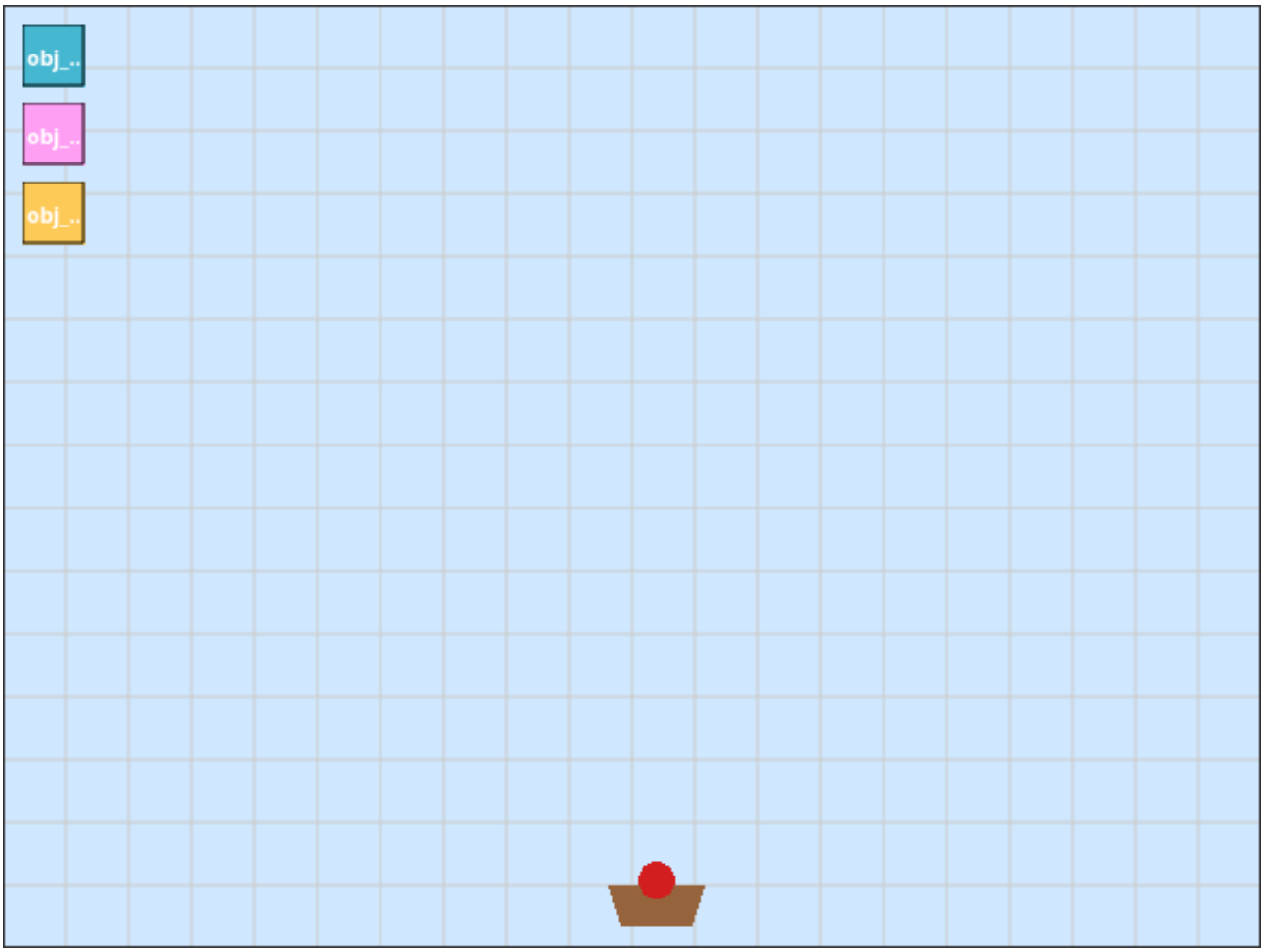
You should see: press **F5**. Catch a cherry, then a strawberry — the basket ends up an orange, having scored 30 points.



Phase 4: Winning

- [ ] **15.** Create a sprite spr_fruit_orange (32x32, an orange circle) and an object obj_fruit_orange, built exactly like the others. There is no falling watermelon sprite or spawner — a watermelon is only ever something your basket becomes.
- [ ] **16.** Create obj_spawn_orange (alarm 0 to 150; random x from 0 to 608, since the orange is 32 wide) and place one in room_main.
- [ ] **17.** Create obj_win_text. **Draw** event: *Draw text* "YOU WIN! You made a watermelon!" at x 120, y 220 and "Press SPACE to play again" at x 190, y 260. **Key press: space**: *Restart game*.
- [ ] **18.** Create room_win with a bright background color, and place one obj_win_text instance in it.
- [ ] **19.** Create a sprite spr_basket_watermelon (56x40, a bowl with a green, watermelon-striped circle in it) — you'll need it in the next step.
- [ ] **20.** Add **Collision with obj_fruit_orange** on obj_player: *If held_level equals 3 then* [*set held_level to 4*, *add 50 to score*, *set sprite to spr_basket_watermelon*, *Go to room room_win*] *Otherwise* [*add 1 to score*]; then *Destroy other instance*.

> **You should see:** press **F5**. Chain all three fusions and the YOU WIN! room appears. SPACE starts a new game.



Bonus (optional): Import Ready-Made Art

Page 6 of the tutorial shows how to swap any of your drawings for ready-made pictures instead, if you'd rather not draw. Your game works exactly the same either way — this is only worth doing once everything above already works.

Stuck?

What is wrong	Most likely cause	What to do
The basket never changes picture	set_sprite is missing from the *If* branch, or it points at the wrong sprite name	Check the exact sprite name in *Set sprite to*
Catching a matching fruit only gives 1 point	held_level is not actually equal to the number the *If* checks, or the condition uses the wrong operator	Check *If held_level equals* uses the right number for that phase (1, then 2, then 3)
The basket is stuck as a cherry forever	The merge's *Otherwise* branch accidentally changed held_level too	Only the *If* branch should touch held_level and the sprite
Fruit piles up and never disappears	The fruit's **Outside Room** event is missing	Add *Destroy this instance* in Outside Room

No fruit ever falls	The spawner object has no instance placed in room_main, or its alarm was never set in Create	Place one spawner instance; check *Set alarm 0* is in **Create**, not somewhere else
Catching an orange does not win	The *Go to room room_win* action is in the wrong place, or the room name is misspelled	It must be inside the *If held_level equals 3* branch, spelled exactly room_win
The win screen is blank	obj_win_text was not placed in room_win, or its Draw event is missing	Place one instance; check the Draw event exists
SPACE does nothing on the win screen	The win room has no object with a **Key press: space** event	Add the event to obj_win_text

Challenges

- **Try this (5 minutes):** change the fruit fall speed, or how often a spawner fires.
- **Push further:** add a fourth spawner and fruit type between orange and watermelon.
- **Invent:** add lives that go down when you mismatch too many times in a row.

Vocabulary

Term	What it means
Held fruit	The one fruit the basket is currently carrying, remembered in held_level
Fusion	Catching a matching fruit so it grows into the next, bigger fruit
If / Else	A block that does one thing when something is true, and a different thing when it is not
Spawner	An invisible object whose only job is to create other objects on a timer
Outside Room	An event that fires when an instance leaves the room — used here to clean up missed fruit

Check Yourself

Write your answers in the notes below.

1. Why does the basket need a variable like held_level at all?
2. What happens if you catch a cherry while already holding an orange?
3. Why is there no sprite or spawner for a falling watermelon?

My Notes